

Name: \_\_\_\_\_

Class: 12MTX\_\_\_\_\_

Teacher: \_\_\_\_\_

**CHERRYBROOK TECHNOLOGY HIGH SCHOOL**



**2005**

**YEAR 12**

**AP4 EXAMINATION**

**MATHEMATICS EXTENSION 1**

*Time allowed - 2 HOURS  
(Plus 5 minutes' reading time)*

**DIRECTIONS TO CANDIDATES:**

- Attempt all questions.
- All questions are of equal value.
- Each question is to be commenced on a new page clearly marked Question 1, Question 2, etc on the top of the page. \*\*
- Each question is to be returned in a separate bundle.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.

***\*\*Each page must show your name and your class. \*\****

**Question 1 (12 marks)****Marks**

- (a) Let  $A$  be the point  $(-3, 8)$  and let  $B$  be the point  $(5, -6)$ . Find the coordinates of the point  $P$  that divides the interval  $AB$  internally in the ratio  $1:3$ . 2

- (b) What is the remainder when the polynomial  $P(x) = x^3 + 3x^2 - 1$  is divided by  $x - 2$ ? 2

- (c) Use the table of standard integrals to find the exact value of 2

$$\int_0^1 \frac{1}{\sqrt{x^2 + 9}} dx.$$

- (d) Solve  $\frac{2}{x+5} \leq 1$ . 3

- (e) Use the substitution  $u = x - 1$  to evaluate  $\int_2^4 \frac{x}{(x-1)^2} dx$ . 3

**Question 2 (12 marks)**

- (a) Sketch the graph of  $y = 2\sin^{-1} 3x$  showing clearly the domain and range of the function as well as any intercepts. 2

- (b) Let  $f(x) = 4x^2 - 1$ . 2  
Use the definition  $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$  to find the derivative of  $f(x)$  at  $x = a$ .

**Question 2 (continued)****Marks**

(c) Find  $\frac{d}{dx}(3x^2 \cos^{-1} x)$  **2**

(d) Find  $\int 4 \cos^2 3x \, dx$ . **2**

(e) Solve the equation  $\sin 2\theta = \sqrt{2} \cos \theta$  for  $0 \leq \theta \leq 2\pi$ . **4**

**Question 3 (12 marks)**

(a) The variable point  $(2\cos\theta, 3\sin\theta)$  lies on a curve. Find the Cartesian equation of this curve. **2**

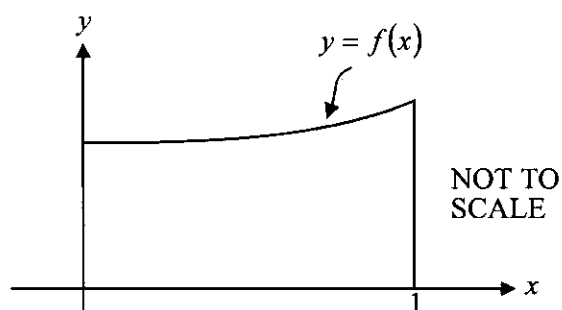
(b) The function  $f(x) = \log_e x + 5x$  has a zero near  $x = 0.2$ . **3**  
Using  $x = 0.2$  as a first approximation, use one application of Newton's method to find a second approximation to the zero.  
Write your answer correct to 3 decimal places.

**Question 3 (continued)**

**Marks**

(c) For the function  $f(x) = \frac{1}{\sqrt{4-x^2}}$

- (i) Find the natural domain of the function. 1
- (ii) The sketch below shows part of the graph of  $y = f(x)$ . The area under the curve for  $0 \leq x \leq 1$  is shaded. Find the area of the shaded region. 2



(d) A particle moves in simple harmonic motion about a fixed point  $O$ . The amplitude of the motion is 2 m and the period is  $\frac{2\pi}{3}$  seconds. Initially the particle moves from  $O$  with a positive velocity.

- (i) Explain why the displacement  $x$ , in metres, of the particle at time  $t$  seconds, can be given by 1
- $$x = 2\sin 3t$$
- (ii) Find the speed of the particle when it is  $\sqrt{3}$  m from  $O$ . 2
- (iii) What is the maximum speed reached by the particle? 1

**Question 4 (12 marks)**

**Marks**

- (a) Use mathematical induction to prove that

**3**

$$1 + 6 + 15 + \dots + n(2n-1) = \frac{1}{6}n(4n-1)(n+1)$$

for all positive integers  $n$ .

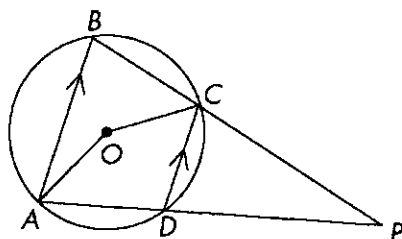
- (b) (i) Show that  $\frac{1}{1 - \tan x} - \frac{1}{1 + \tan x} = \tan 2x$

**2**

- (ii) Evaluate  $\frac{1}{1 - \tan \frac{\pi}{6}} - \frac{1}{1 + \tan \frac{\pi}{6}}$  in simplest exact form

**1**

- (c) In the diagram below,  $O$  is the centre of the circle and  $AB \parallel DC$ .  $AD$  and  $BC$  meet at  $P$ .



Prove: (i)  $CP = DP$ .

**2**

(ii)  $\triangle ABP$  is isosceles.

**2**

(iii)  $OAPC$  is a cyclic quadrilateral.

**2**

**Question 5 (12 marks)****Marks**

- (a) Solve for  $x$

$$x^{\log_2 x} = 8x^2 \ (x > 0)$$

**3**

- (b) Consider the function  $f(x) = x(x-2)^2$ ,  $x \leq a$  where  $a$  is a constant.

- (i) Find the values of  $a$  given that the inverse function  $f^{-1}(x)$  of  $f(x)$  exists.

**2**

- (ii) State the domain of  $f^{-1}(x)$ .

**1**

- (c) If  $\alpha$ ,  $\beta$  and  $\gamma$  are the roots of the cubic equation  $x^3 - 4x^2 + 3x + 2 = 0$ , find  $\alpha^2 + \beta^2 + \gamma^2$ .

**2**

- (d) Factorise  $m^3 - 3m + 2$  and solve the equation  $(3x-4)^3 - 9x + 14 = 0$ .

**4**

**Question 6 (12 marks)****Marks**

- (a) A particle moves in a straight line with an acceleration given by

$$\frac{d^2x}{dt^2} = 9(x - 2)$$

where  $x$  is the displacement in metres from an origin  $O$  after  $t$  seconds. Initially, the particle is 4 metres to the right of  $O$  so that  $x = 4$  and has velocity  $v = -6$ .

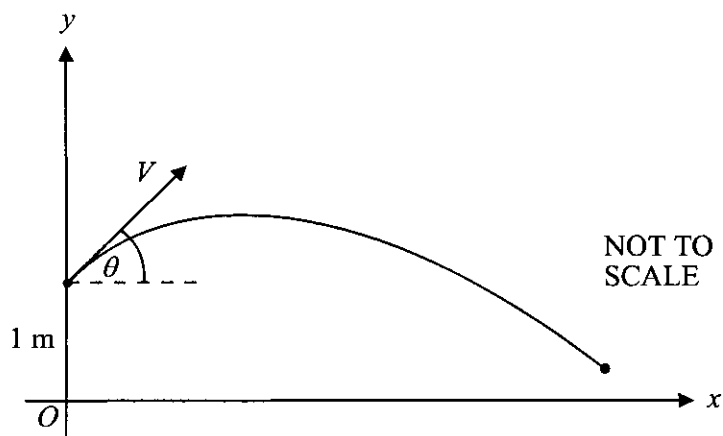
- |       |                                                                      |   |
|-------|----------------------------------------------------------------------|---|
| (i)   | Show that $v^2 = 9(x - 2)^2$ .                                       | 2 |
| (ii)  | Find an expression for $v$ and hence find $x$ as a function of $t$ . | 2 |
| (iii) | Explain whether the velocity of the particle is ever zero.           | 2 |

*Question 6 continues on the next page.*

**Question 6 (continued)**

**Marks**

- (b) A boy throws a ball and projects it with a speed of  $V \text{ m s}^{-1}$  from a point 1 metre above the ground. The ball lands on top of a flowerpot in a neighbour's yard.



The angle of projection is  $\theta$  as indicated in the diagram.  
The equations of motion of the ball are

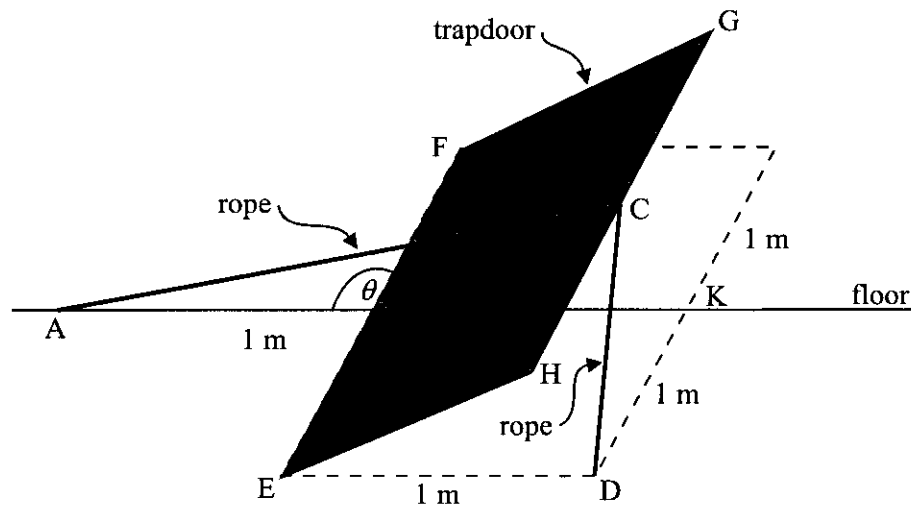
$$\ddot{x} = 0 \text{ and } \ddot{y} = -10$$

where  $x$  and  $y$  are shown on the axes on the diagram. The position of the ball  $t$  seconds after it is thrown by the boy is described by the coordinates  $(x, y)$ .

It has been found that  $y = Vt \sin \theta - 5t^2 + 1$ .

- |       |                                                                                                                                                                                                   |          |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| (i)   | Show that $x = Vt \cos \theta$ .                                                                                                                                                                  | <b>2</b> |
| (ii)  | When the ball is at its maximum height above the ground, it passes directly above a 1.5 metre high fence and clears the fence by 0.5 metres.<br>Find an expression for $V$ in terms of $\theta$ . | <b>2</b> |
| (iii) | Find the value of $V$ given that $\theta = \tan^{-1} \frac{9}{40}$ .<br>Give your answer in $\text{m s}^{-1}$ , correct to 2 decimal places.                                                      | <b>2</b> |

(a)



A rectangular trapdoor is shown in the diagram as  $EFGH$  where  $EH = 1$  m,  $EF = 2$  m and  $BC$  divides the trapdoor in half.

A rope is anchored at point  $A$  on the floor 1 metre from point  $B$  and points  $A$ ,  $B$  and  $K$  lie on a straight line.

The rope passes through a small loop on the edge of the trapdoor at point  $C$  and is anchored to the floor at point  $D$ .

As the trapdoor is being opened or closed, the rope running from  $A$  through  $C$  to  $D$  is kept taut by pulling it tight or letting it out through anchor point  $A$ .

Let  $\angle ABC = \theta$ ,  $0^\circ \leq \theta \leq 180^\circ$ .

- (i) Show that  $AC = \sqrt{2 - 2\cos\theta}$ . 1
- (ii) Show that  $CD = \sqrt{3 + 2\cos\theta}$ . 2
- (iii) Let  $l$  equal the length of the rope from  $A$  through  $C$  to  $D$ . find the maximum value of  $l$ . Justify your answer. 4

**Question 7 (continued)****Marks**

- (b) A cup of soup with a temperature  $95^{\circ}\text{C}$  is placed in a room which has a temperature of  $20^{\circ}\text{C}$ . In 10 minutes the cup of soup cools to  $70^{\circ}\text{C}$ . Assuming the rate of heat loss is proportional to the excess of its temperature above room temperature, that is

$$\frac{dT}{dt} = -k(T - 20),$$

- (i) show that  $T = 20 + Ae^{-kt}$  is a solution of

**1**

$$\frac{dT}{dt} = -k(T - 20).$$

- (ii) find the temperature of the soup after a further 5 min. to the nearest degree.

**2**

- (iii) how long will it take the soup to cool to  $35^{\circ}\text{C}$ ?  
Give your answer correct to the nearest minute.

**1**

- (iv) find the rate of cooling when the soup is  $35^{\circ}\text{C}$ .  
Give your answer correct to 1 decimal place.

**1**

①

# TRIAL SOLUTIONS

## AP4 EXT 1 2005

### Question 1

$$a) P = \left( \frac{1 \times 5 + 3 \times -3}{4}, \frac{1 \times -6 + 3 \times 8}{4} \right) \textcircled{1}$$

$$= (-1, 4\frac{1}{2}) \textcircled{1}$$

$$b) P(x) = x^3 + 3x^2 - 1$$

using remainder theorem

$$P(2) = 8 + 12 - 1 \textcircled{1}$$

$$= 19 \textcircled{1}$$

there is remainder 19.

$$c) \int_0^1 \frac{1}{\sqrt{x^2+9}} dx = \left[ \ln(x + \sqrt{x^2+9}) \right]_0^1$$

$$= \ln(1 + \sqrt{10}) - \ln(0+3) \textcircled{1}$$

$$= \ln\left(\frac{1+\sqrt{10}}{3}\right) \textcircled{1}$$

$$d) \frac{2}{(x+5)} \leq 1 \quad x \neq -5$$

$$2(x+5) \leq (x+5)^2 \textcircled{1}$$

$$2x+10 \leq x^2+10x+25$$

$$0 \leq x^2+8x+15$$

$$0 \leq (x+5)(x+3) \textcircled{1}$$

$$x \leq -5 \text{ or } x \geq -3$$

$$\text{but } x \neq -5$$

$$x < -5 \text{ or } x \geq -3 \textcircled{1}$$

$$e) \int_2^4 \frac{x}{(x-1)^2} = \int_1^3 (u+1)u^{-2} du \textcircled{1}$$

$$= \int_1^3 (u^{-1} + u^{-2}) du$$

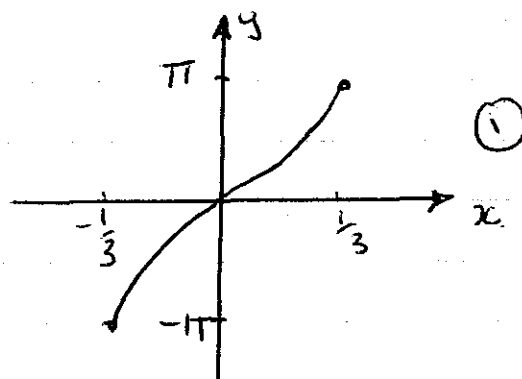
$$= \left[ \log_e u - u^{-1} \right]_1^3 \textcircled{1}$$

$$= \left( \log_e 3 - \frac{1}{3} \right) - (\log_e 1 - 1)$$

$$= \log_e 3 + \frac{2}{3} \textcircled{1}$$

### Question 2

$$a) \begin{cases} -\pi \leq y \leq \pi \\ -\frac{1}{3} \leq x \leq \frac{1}{3} \end{cases} \textcircled{1} \text{ for either}$$



$$b) f'(a) = \lim_{h \rightarrow 0} \frac{4(a+h)^2 - 1 - [4a^2 - 1]}{h} \textcircled{1}$$

$$= \lim_{h \rightarrow 0} \frac{4a^2 + 8ah + 4h^2 - 1 - 4a^2 + 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{8ah + 4h^2}{h}$$

$$= \lim_{h \rightarrow 0} 8a + 4h$$

$$= 8a \textcircled{1}$$

Question 2 (continued)

c)  $\frac{d}{dx}(3x^2 \cos^{-1} x)$

$$= 6x \cos^{-1} x - \frac{3x^2}{\sqrt{1-x^2}} \quad (1)$$

$$u = 3x^2 \quad v = \cos^{-1} x$$

$$u' = 6x \quad v' = \frac{-1}{\sqrt{1-x^2}} \quad (1)$$

d)  $\int 4 \cos^2 3x \, dx$

$$= 4 \int \left( \frac{1}{2} + \frac{1}{2} \cos 6x \right) dx \quad (1)$$

$$\text{since } \cos^2 \theta = \frac{1}{2} + \frac{1}{2} \cos 2\theta$$

$$= 4 \left[ \frac{x}{2} + \frac{1}{12} \sin 6x \right] + C$$

$$= 2x + \frac{1}{3} \sin 6x + C \quad (1)$$

e)  $\sin 2\theta = \sqrt{2} \cos \theta$

$$\Rightarrow 2 \sin \theta \cos \theta = \sqrt{2} \cos \theta \quad (1)$$

$$\sqrt{2} \sin \theta \cos \theta - \cos \theta = 0$$

$$\cos \theta (\sqrt{2} \sin \theta - 1) = 0 \quad (1)$$

$$\therefore \cos \theta = 0 \quad \text{or} \quad \sin \theta = \frac{1}{\sqrt{2}}$$

(1)

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}$$

(1)

$$\therefore \theta = \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{3\pi}{2}$$

Question 3

a)  $x = 2 \cos \theta \Rightarrow \cos \theta = \frac{x}{2} \quad (1)$

$$y = 3 \sin \theta \Rightarrow \sin \theta = \frac{y}{3}$$

$$\text{since } \sin^2 \theta + \cos^2 \theta = 1$$

$$\therefore \frac{x^2}{4} + \frac{y^2}{9} = 1 \quad (1)$$

b)  $f(x) = \log_e x + 5x$

$$f'(x) = \frac{1}{x} + 5$$

$$x_0 = 0.2$$

$$\text{Now } x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \quad (1)$$

$$= 0.2 - \frac{\log_e 0.2 + 5 \times 0.2}{\frac{1}{0.2} + 5} \quad (1)$$

$$= 0.261 \quad (\text{correct to } 3 \text{ dec pl.})$$

c) i)  $4 - x^2 \geq 0$  so  $-2 \leq x \leq 2$

$$\text{but } \sqrt{4-x^2} \neq 0 \therefore x \neq -2, 2$$

 $\therefore$  Natural domain is

$$-2 < x < 2$$

## Question 3 (continued)

c) ii)

$$A = \int_0^1 \frac{1}{\sqrt{4-x^2}} dx$$

$$= \left[ \sin^{-1}\left(\frac{x}{2}\right) \right]_0^1 \quad (1)$$

$$= \sin^{-1} \frac{1}{2} - \sin^{-1} 0$$

$$= \frac{\pi}{6} \quad (1)$$

$$\therefore \text{Area} = \frac{\pi}{6} \text{ units}^2.$$

d) Particle starts from centre of motion with positive velocity (i.e.  $x=0$ .)

$\therefore$  general form of displacement time function is  $x = a \sin nt$ .

$$\text{Now } a=2, \text{ period} = \frac{2\pi}{n} = \frac{2\pi}{3}$$

$$\therefore n=3.$$

$\therefore$  required equation is

$$x = 2 \sin 3t.$$

(1) for correctly derived eqn.

ii) ~~3~~

$$v^2 = n^2(a^2 - x^2) \\ = 9(4 - x^2)$$

when  $x = \sqrt{3}$

$$v^2 = 9(4 - 3) \quad (1) \\ = 9$$

$$\therefore v = \pm 3$$

initially positive velocity:

$$\therefore v = 3 \text{ m/s} \quad (1)$$

iii)

For SHM max speed occurs at centre of motion i.e.  $x=0$

$$\therefore v^2 = n^2(a^2 - x^2) \\ = 9(4 - 0) \\ = 36$$

$$\therefore v = \pm 6 \text{ m/s}$$

$\therefore$  Maximum speed is 6 m/s. (1)

### Question 4

a) For  $n=1$

$$LHS = 1$$

$$\begin{aligned} RHS &= \frac{1}{6} [(4-1)(1+1)] \\ &= \frac{1}{6} (3 \times 2) \\ &= 1. \end{aligned}$$

$\therefore$  true for  $n=1$  (1)

Assume true for  $n=k$ .

$$\begin{aligned} \therefore 1+6+15+\dots+k(2k-1) \\ = \frac{1}{6} k(4k-1)(k+1) \end{aligned}$$

For  $n=k+1$

$$\begin{aligned} 1+6+15+\dots+k(2k-1)+(k+1)(2k+1) \\ = \frac{1}{6} k(4k-1)(k+1) + (k+1)(2k+1) \end{aligned} \quad (1) \quad c)$$

$$= \frac{1}{6} (k+1) [k(4k-1) + 6(2k+1)]$$

$$= \frac{1}{6} (k+1) [4k^2 - k + 12k + 6]$$

$$= \frac{1}{6} (k+1) [4k^2 + 11k + 6]$$

$$= \frac{1}{6} (k+1) (4k+3)(k+2)$$

$\therefore$  If true for  $n=k$  it is

true for  $n=k+1$

but true for  $n=1$ ,  $\therefore$  true for  $n=2$  and hence 3, 4, 5...

$\therefore$  true for all integers

$n \geq 1$ . (1)

$$b) i) \frac{1}{1-\tan x} - \frac{1}{1+\tan x}$$

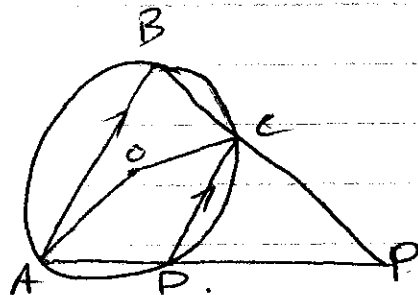
$$= \frac{1+\tan x - (1-\tan x)}{1-\tan^2 x} \quad (1)$$

$$= \frac{2 \tan x}{1-\tan^2 x} \quad (1)$$

$$= \tan 2x.$$

$$ii) \frac{1}{1-\tan \frac{\pi}{6}} - \frac{1}{1+\tan \frac{\pi}{6}} = \tan \frac{\pi}{3}$$

$$= \sqrt{3}. \quad (1)$$



$$i) \angle PCD = \angle ABC. \quad (1)$$

(Corresp.  $\angle$ 's;  $AB \parallel DC$ )

$$\angle PDC = \angle ABC \quad (1)$$

(Exterior  $\angle$  of cyclic quad)

$$\therefore \angle PCD = \angle PDC$$

$$\therefore \triangle PCD \text{ is isosceles.}$$

$$\therefore PC = PD.$$

$$ii) \angle PCD = \angle BAD \quad (1)$$

(Corresp.  $\angle$ 's;  $AB \parallel DC$ )

$$\angle PCD = \angle ABC \quad (\text{Exterior angle of cyclic quad})$$

$$\therefore \angle BAD = \angle ABC \quad (1) \text{ (Axiom)}$$

$$\therefore \triangle ABP \text{ is isosceles}$$

(5)

Question 4 (continued)

iii)  $\angle AOC = 2 \angle ABC$  (1)  
 ( $\angle$  at centre twice angle at circumference).  
 $= \angle PDC + \angle PCD$

but  $\angle PDC + \angle PCD + \angle CPD = 180^\circ$   
 ( $\angle$ s of a  $\Delta = 180^\circ$ )

$\therefore \angle AOC + \angle CPD = 180^\circ$  (1)

$\therefore OAPC$  is a cyclic Quad.  
 (opp  $\angle$ s supplementary)

Question 5:

a) ~~log~~  
 $x^{\log_2 x} = 8x^2$

take  $\log_2$  on both sides

$\log_2 (x^{\log_2 x}) = \log_2 8x^2$  (1)

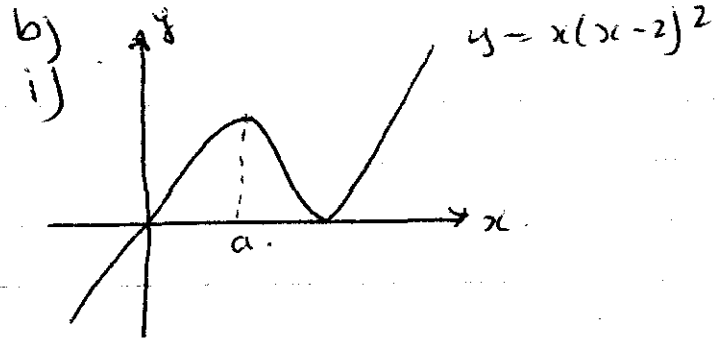
$\log_2 x \times \log_2 x = \log_2 8 + \log_2 x^2$

$(\log_2 x)^2 - 2 \log_2 x - 3 = 0$

$(\log_2 x - 3)(\log_2 x + 1) = 0$  (1)

$\therefore \log_2 x = 3$  or  $\log_2 x = -1$

$\therefore x = 8$  or  $x = \frac{1}{2}$  (1)



The inverse will exist if  
 $x \leq a$  where  $a$  is the turning  
 pt.

$y = 3x^2 - 8x + 4$  (1)  
 $= (3x-2)(x-2)$

$y' = 0$  if

$3x-2=0$  or  $x-2=0$

$x = \frac{2}{3}$

$x = 2$

$\therefore f^{-1}(x)$  exists if  $a \leq \frac{2}{3}$  (1)

ii) Domain of  $f^{-1}(x)$  is

$x \leq \left(\frac{2}{3} - 2\right)^2 \left(\frac{2}{3}\right)$

$\therefore x \leq \frac{32}{27}$  (1)

6

Question 5 (continued)

c)  $\alpha + \beta + \gamma = -\frac{b}{a} = 4$

$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} = 3$  ①

$\alpha\beta\gamma = -\frac{d}{a} = -2$

$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$   
 $= 16 - 2(3)$   
 $= 10$  ①

d) let  $f(m) = m^3 - 3m + 2$

$f(1) = 1 - 3 + 2 = 0$

$\therefore m-1$  is a factor ①

$$\begin{array}{r} m^2 + m - 2 \\ m-1 \overline{) m^3 - 3m + 2} \\ \underline{m^3 - m^2} \phantom{+ 2} \\ m^2 - 3m \phantom{+ 2} \\ \underline{m^2 - m} \phantom{+ 2} \\ -2m + 2 \\ \underline{-2m + 2} \phantom{+ 2} \end{array}$$

$\therefore$  Factors are  $(m-1)(m^2 + m - 2)$   
 i.e.  $(m-1)(m-1)(m+2)$  ①

$(3x-4)^3 - 3(2x-5) - (3x+1)$   
 $= (3x-4)^3 - 6x + 15 - 3x - 1$   
 $= (3x-4)^3 - 9x + 14$   
 $= (3x-4)^3 - 3(3x-4) + 2$

$\therefore (3x-4)^3 - 3(3x-4) + 2 = 0$  ①

let  $m = 3x-4$

$\therefore (m-1)^2(m+2) = 0$

$\therefore (m-1)^2 = 0$  or  $m+2=0$

$\therefore x = \frac{5}{3}$  or  $x = \frac{2}{3}$  ①

Question 6

a) (i)  $\frac{d^2x}{dt^2} = 9(x-2)$

$\frac{d}{dx} \frac{1}{2} v^2 = 9(x-2)$

$\therefore \frac{1}{2} v^2 = 9 \int (x-2) dx$

$= 9 \left( \frac{x^2}{2} - 2x \right) + C$

initially  $x=4, v=-6$

$\therefore 18 = 9(8-8) + C$

$\therefore C = 18$  ①

$\therefore \frac{1}{2} v^2 = 9 \left( \frac{x^2}{2} - 2x \right) + 18$

$v^2 = 9x^2 - 36x + 36$   
 $= 9(x^2 - 4x + 4)$   
 $= 9(x-2)^2$  ①

as required.

Question 6 (continued)

$$\text{ii) } v^2 = 9(x-2)^2$$

$$v = \pm 3(x-2)$$

$$t=0, x=4 \quad v=-6$$

$$\therefore v = -3(x-2) \quad \textcircled{1}$$

$$\text{so } \frac{dx}{dt} = -3(x-2)$$

$$\frac{dt}{dx} = \frac{-1}{3(x-2)}$$

$$t = -\frac{1}{3} \int \frac{1}{x-2} dx$$

$$= -\frac{1}{3} \log_e(x-2) + C$$

$$t=0 \quad x=4$$

$$\therefore 0 = -\frac{1}{3} \log_e 2 + C$$

$$\therefore C = \frac{1}{3} \log_e 2$$

$$\therefore t = \frac{1}{3} \log_e \frac{2}{x-2}$$

$$3t = \log_e \frac{2}{x-2}$$

$$e^{3t} = \frac{2}{x-2}$$

$$x-2 = 2e^{-3t}$$

$$x = 2(1 + e^{-3t}) \quad \textcircled{1}$$

$$\text{iii) } v = -3(x-2)$$

$$\text{if } v=0 \quad x=2 \quad \textcircled{1}$$

$$\text{if } x=2 \quad e^{-3t} = 0 \text{ which}$$

$$\text{has no solution} \quad \textcircled{1}$$

$$\therefore v \text{ is never } 0.$$

$$\text{b) i) } \ddot{x} = 0$$

$$\dot{x} = \int 0 dt$$

$$= C$$

$$\text{when } t=0, \dot{x} = v \cos \theta$$

$$\therefore \dot{x} = v \cos \theta \quad \textcircled{1}$$

$$x = \int v \cos \theta dt$$

$$= vt \cos \theta + C_1$$

$$\text{when } t=0, x=0$$

$$\therefore C_1 = 0 \quad \textcircled{1}$$

$$\therefore x = vt \cos \theta \text{ as required.}$$

$$\text{ii) Max. height} = 2 \text{ if}$$

$$\text{occurs when } \dot{y} = 0.$$

$$y = vt \sin \theta - 5t^2 + 1$$

$$\dot{y} = v \sin \theta - 10t$$

$$\text{if } \dot{y}=0$$

$$0 = v \sin \theta - 10t$$

$$t = \frac{v \sin \theta}{10} \quad \textcircled{1}$$

Question 6 b (continued)

Now  $y = vt \sin \theta - 5t^2 + 1$

$$\therefore 2 = \frac{v^2 \sin^2 \theta}{10} - \frac{5v^2 \sin^2 \theta}{100} + 1$$

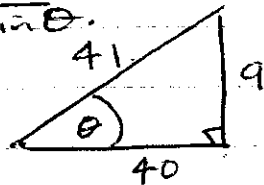
$$1 = \frac{v^2 \sin^2 \theta}{20}$$

$$v^2 = \frac{20}{\sin^2 \theta}$$

$$v = \frac{\sqrt{20}}{\sin \theta} \quad (1) \quad (v > 0)$$

$$= \frac{2\sqrt{5}}{\sin \theta}$$

iii)



$$v = \frac{\sqrt{20}}{\sin \theta}$$

also  $\theta = \tan^{-1} \frac{9}{40}$

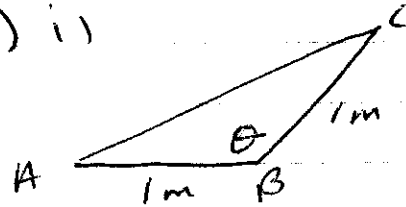
$$\therefore \sin \theta = \frac{9}{41} \quad (1)$$

$$\therefore v = \frac{2\sqrt{5} \times 41}{9}$$

$$= 20.37 \text{ m/s}^{-1} \quad (1)$$

Question 7

a) i)

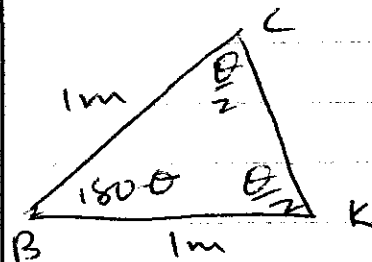


$$\therefore AC^2 = 1^2 + 1^2 - 2 \times 1 \times 1 \cos \theta \quad (1)$$

$$= 2 - 2 \cos \theta$$

$$\therefore AC = \sqrt{2 - 2 \cos \theta} \text{ as required}$$

ii) In  $\triangle BCK$



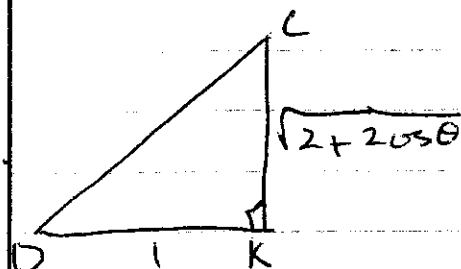
$$\therefore CK^2 = 1^2 + 1^2 - 2 \times 1 \times 1 \cos(180 - \theta)$$

$$= 2 - 2 \cos(180 - \theta)$$

$$= 2 + 2 \cos \theta$$

$$CK = \sqrt{2 + 2 \cos \theta} \quad (1)$$

In  $\triangle CDK$



$$\therefore CD^2 = 1^2 + (\sqrt{2 + 2 \cos \theta})^2 \quad (1)$$

$$= 1 + 2 + 2 \cos \theta$$

$$= 3 + 2 \cos \theta$$

$$CD = \sqrt{3 + 2 \cos \theta} \text{ as required}$$

(9)

Question 7 (continued)

$$\text{iii) } L = AC + CD \\ = \sqrt{2-2\cos\theta} + \sqrt{3+2\cos\theta}$$

$$\text{so } \frac{dL}{d\theta} = \frac{1}{2}(2-2\cos\theta)^{-\frac{1}{2}} \times 2\sin\theta$$

$$+ \frac{1}{2}(3+2\cos\theta)^{-\frac{1}{2}} \times -2\sin\theta \text{ when } \theta = \cos^{-1}\left(-\frac{1}{4}\right)$$

$$= \frac{\sin\theta}{\sqrt{2-2\cos\theta}} - \frac{\sin\theta}{\sqrt{3+2\cos\theta}} \quad (1)$$

we require

$$\frac{dL}{d\theta} = 0.$$

$$\frac{dL}{d\theta} = \frac{\sin\theta \sqrt{3+2\cos\theta} - \sin\theta \sqrt{2-2\cos\theta}}{(\sqrt{2-2\cos\theta})(\sqrt{3+2\cos\theta})}$$

$$\frac{dL}{d\theta} = 0 \text{ when}$$

$$\sin\theta(\sqrt{3+2\cos\theta} - \sqrt{2-2\cos\theta}) = 0$$

$$\therefore \sin\theta = 0 \text{ or } \sqrt{3+2\cos\theta} = \sqrt{2-2\cos\theta}$$

$$\theta = 0^\circ \text{ or } 180^\circ \text{ or } 3+2\cos\theta = 2-2\cos\theta$$

$$1 = -4\cos\theta$$

$$(1) \quad \cos\theta = -\frac{1}{4} \\ \theta = \cos^{-1}\left(-\frac{1}{4}\right)$$

When  $\theta = 0^\circ$  trap door fully open  $\therefore$  distance =  $\sqrt{5}$ .

If  $\theta = 180^\circ$  trap door is closed and  $L = AK + KD$

$$= 2 + 1$$

$$= 3\text{m}$$

These two are end points of function.

$$\text{when } \theta = \cos^{-1}\left(-\frac{1}{4}\right)$$

$$L = \sqrt{2-2\cos\left(\cos^{-1}\left(-\frac{1}{4}\right)\right)} +$$

$$\sqrt{3+2\cos\left(\cos^{-1}\left(-\frac{1}{4}\right)\right)}$$

$$= \sqrt{2-2\left(-\frac{1}{4}\right)} + \sqrt{3+2\left(-\frac{1}{4}\right)}$$

$$= \sqrt{\frac{5}{2}} + \sqrt{\frac{5}{2}}$$

$$= 2\sqrt{\frac{5}{2}}$$

$$= \sqrt{10} \quad (1)$$

$\therefore$  max length of  $L$  is  $\sqrt{10}$ .  
check for maximum

$$\cos^{-1}\left(-\frac{1}{4}\right) \approx 1.8234 \text{ rad}$$

$$\theta = 1 \quad \frac{dL}{d\theta} \approx 0.4670$$

$$\theta = 2 \quad \frac{dL}{d\theta} \approx 0.0840$$

$\therefore$  change of sign  $\therefore$  maximum. (1)

Question 7 (continued)

a) (i)  $T = 20 + Ae^{-kt}$

$$\frac{dT}{dt} = -kAe^{-kt}$$

$$= -k(T-20) \quad (1)$$

(ii) when  $t=0$ ,  $T=95$

$$95 = 20 + A$$

$$A = 75$$

$$T = 20 + 75e^{-kt}$$

if  $t=10$ ,  $T=70$

$$70 = 20 + 75e^{-10k}$$

$$e^{-10k} = \frac{2}{3}$$

$$\therefore k = -\frac{1}{10} \ln \frac{2}{3} \quad (1)$$

$\therefore$  if  $t=15$

$$T = 20 + 75e^{\frac{1}{10} \ln \frac{2}{3} (15)}$$

$$= 60.82 \dots \quad (1)$$

$$= 61^\circ \text{C.} \quad \checkmark$$

iii)  $35 = 20 + 75e^{\frac{1}{10} \ln \frac{2}{3} t}$

$$t = \frac{\ln \frac{1}{5}}{\frac{1}{10} \ln \frac{2}{3}}$$

$$= 39.69 \dots \quad (1)$$

$$= 40 \text{ min.}$$

iv)  $\frac{dT}{dt} = 75 \times \frac{1}{10} \ln \frac{2}{3} e^{\frac{1}{10} \ln \frac{2}{3} t}$

$t=40. \therefore \frac{dT}{dt} = -0.600$

$$= -0.6 \text{ deg/min} \quad (1)$$